THE CECH COMPLEX OF A FAMILY OF MORPHISMS IS AN EXACT FUNCTOR ON ABELIAN PRESHEAVES

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument: the pointwise criterion is stated, the degreewise product structure is identified, and the conclusion follows; the closing pointer to the same statement in another chapter is a reference and not a substitution for the argument.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter etale-cohomology.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- The Cech complex of a family of morphisms. Same chapter, Definition definition-cech-complex, recorded as the context of this card. Supplies the complex whose exactness is asserted.
- Abelian presheaves form a functor category. Standard: the category of abelian presheaves on a category is the category of functors to abelian groups, hence abelian with exactness checked pointwise. Turns exactness of a sequence of presheaves into exactness of the sequences of values at every object.
- The same statement in the chapter on cohomology on sites. Cohomology on Sites, Lemma sites-cohomology-lemma-cech-exact-presheaves. Is the pointer the author gives for the same computation.

Source reference. Original lemma 03AQ: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 1530–1530. Navigation pointer invoked at proof line 1530; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:etale-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Definition 0.1. Let $\mathcal{C}$ be a category. Let $\mathcal{U} = \{U_i \to U\}_{i \in I}$ be a family of morphisms of $\mathcal{C}$ with fixed target. Assume that all the fibre products $U_{i_0} \times_U \ldots \times_U U_{i_p}$ exists in $\mathcal{C}$. Let $\mathcal{F} \in PAb(\mathcal{C})$ be an abelian presheaf. We define the *Čech complex* $\check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{F})$ by

$$\prod_{i_0 \in I} \mathcal{F}(U_{i_0}) \to \prod_{i_0, i_1 \in I} \mathcal{F}(U_{i_0} \times_U U_{i_1}) \to \prod_{i_0, i_1, i_2 \in I} \mathcal{F}(U_{i_0} \times_U U_{i_1} \times_U U_{i_2}) \to \ldots$$

where the first term is in degree 0 and the maps are the usual ones. The *Čech cohomology groups* are defined by

$$\check{H}^p(\mathcal{U}, \mathcal{F}) = H^p(\check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{F})).$$

Lemma 0.2. *Notation and assumptions as in Definition 0.1. The functor $\check{\mathcal{C}}^\bullet(\mathcal{U}, -)$ is exact on the category $PAb(\mathcal{C})$.*

Proof. This follows at once from the definition of a short exact sequence of presheaves. Namely, as the category of abelian presheaves is the category of functors on some category with values in Ab, it is automatically an abelian category: a sequence $\mathcal{F}_1 \to \mathcal{F}_2 \to \mathcal{F}_3$ is exact in PAb if and only if for all $U \in \mathrm{Ob}(\mathcal{C})$, the sequence $\mathcal{F}_1(U) \to \mathcal{F}_2(U) \to \mathcal{F}_3(U)$ is exact in Ab. So the complex above is merely a product of short exact sequences in each degree. See also Cohomology on Sites, Lemma 03AQ. □